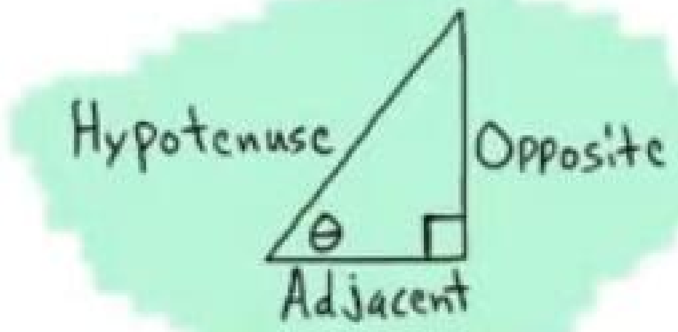
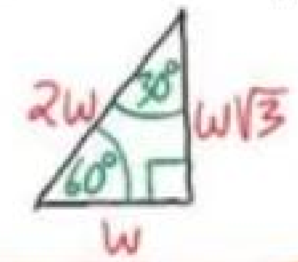


$\sin\theta = \frac{Opp}{Hyp}$ $\csc\theta = \frac{Hyp}{Opp}$

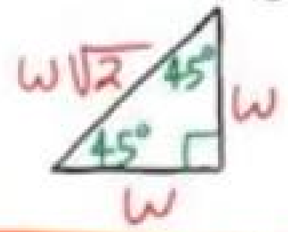
$\cos\theta = \frac{Adj}{Hyp}$ $\sec\theta = \frac{Hyp}{Adj}$

$\tan\theta = \frac{Opp}{Adj}$ $\cot\theta = \frac{Adj}{Opp}$

30°-60°-90° Triangle Theorem: If the inner angles of a triangle are 30°, 60°, and 90°, and the length of the side opposite the 30° angle is w, then the length of the hypotenuse is 2w and the length of the side opposite the 60° angle is $w\sqrt{3}$.



45°-45°-90° Triangle Theorem: If the inner angles of a triangle are 45°, 45°, and 90°, and the length of the two sides opposite the 45° angles are w, then the length of the hypotenuse is $w\sqrt{2}$.



Reciprocal Identities

$\sin\theta = \frac{1}{\csc\theta}$ $\csc\theta = \frac{1}{\sin\theta}$

$\cos\theta = \frac{1}{\sec\theta}$ $\sec\theta = \frac{1}{\cos\theta}$

$\tan\theta = \frac{1}{\cot\theta}$ $\cot\theta = \frac{1}{\tan\theta}$

Ratio Identities

$\tan\theta = \frac{\sin\theta}{\cos\theta}$

$\cot\theta = \frac{\cos\theta}{\sin\theta}$

Pythagorean Identities

$\sin^2\theta + \cos^2\theta = 1$

$\tan^2\theta + 1 = \sec^2\theta$ $1 + \cot^2\theta = \csc^2\theta$

Sum and Difference Formulas

$\sin(A \pm B) = \sin A\cos B \pm \sin B\cos A$

$\cos(A \pm B) = \cos A\cos B \mp \sin A\sin B$

Double Angle Formulas

$\sin(2A) = 2\sin A\cos A$

$\cos(2A) = \cos^2 A - \sin^2 A$

$\cos(2A) = 1 - 2\sin^2 A$

$\cos(2A) = 2\cos^2 A - 1$

Half Angle Formulas

$\sin\left(\frac{A}{2}\right) = \pm\sqrt{\frac{1-\cos A}{2}}$

$\cos\left(\frac{A}{2}\right) = \pm\sqrt{\frac{1+\cos A}{2}}$

General Forms of Trigonometric Functions

Sinusoidal functions: $y = A\sin(wx - \phi) + B$, $y = A\cos(wx - \phi) + B$

$y = A\csc(wx - \phi) + B$ $y = A\tan(wx - \phi) + B$

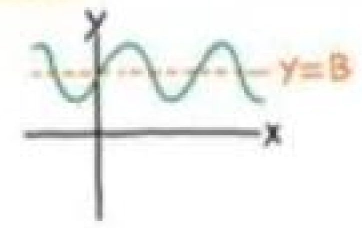
$y = A\sec(wx - \phi) + B$ $y = A\cot(wx - \phi) + B$

Sinusoidal Function: A function whose graph has the same general form as the sine or cosine functions.

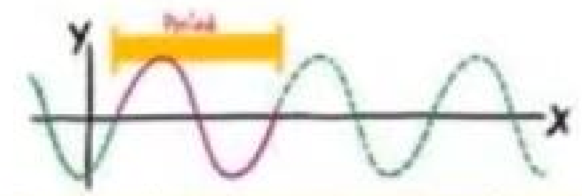
Midline: A horizontal line drawn through the middle of the graph of a trigonometric function to assist when graphing.

For the functions above, the equation of the midline is $y = B$.

Example:



Period: The number of degrees required for a trigonometric function to complete one whole cycle.



The period of the sine, cosine, cosecant, and secant functions is given by the following equation.

$P = \frac{360°}{w}$

The period of the tangent and cotangent functions is given by the following equation.

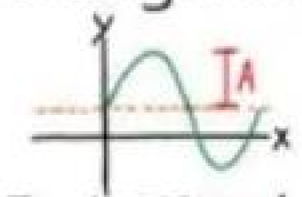
$P = \frac{180°}{w}$

Phase Shift: The horizontal distance that a parent graph is translated (to the right or left).

The phase shift is given by the equation below.

$PS = \frac{\phi}{w}$

Amplitude: The vertical distance from the midline to the highest or lowest point on a sinusoidal graph.



The amplitude is the magnitude (i.e. the absolute value) of the coefficient of the trigonometric function.

Example: If $y = A\sin(wx - \phi) + B$, then Amplitude $= |A|$.

Opposite Angle Identities

$\sin(-\theta) = -\sin\theta$ $\cos(-\theta) = \cos\theta$ $\tan(-\theta) = -\tan\theta$

$\csc(-\theta) = -\csc\theta$ $\sec(-\theta) = \sec\theta$ $\cot(-\theta) = -\cot\theta$

Complete Rotation and Half Rotation Identities

$\sin(\theta \pm 360°) = \sin\theta$ $\sin(\theta \pm 180°) = -\sin\theta$

$\cos(\theta \pm 360°) = \cos\theta$ $\cos(\theta \pm 180°) = -\cos\theta$

$\tan(\theta \pm 360°) = \tan\theta$ $\csc(\theta \pm 180°) = -\csc\theta$

$\csc(\theta \pm 360°) = \csc\theta$ $\sec(\theta \pm 180°) = -\sec\theta$

$\sec(\theta \pm 360°) = \sec\theta$ $\tan(\theta \pm 180°) = \tan\theta$

$\cot(\theta \pm 360°) = \cot\theta$ $\cot(\theta \pm 180°) = \cot\theta$

Complete Rotation Angle Pair Identities and Supplementary Angle Identities

$\sin(360° - \theta) = -\sin\theta$ $\sin(180° - \theta) = \sin\theta$

$\cos(360° - \theta) = \cos\theta$ $\csc(180° - \theta) = \csc\theta$

$\tan(360° - \theta) = -\tan\theta$ $\cos(180° - \theta) = -\cos\theta$

$\csc(360° - \theta) = -\csc\theta$ $\sec(180° - \theta) = -\sec\theta$

$\sec(360° - \theta) = \sec\theta$ $\tan(180° - \theta) = -\tan\theta$

$\cot(360° - \theta) = -\cot\theta$ $\cot(180° - \theta) = -\cot\theta$

Quarter Rotation Identities

$\sin(\theta \pm 90°) = \pm\cos\theta$ $\cos(\theta \pm 90°) = \mp\sin\theta$

$\csc(\theta \pm 90°) = \pm\sec\theta$ $\sec(\theta \pm 90°) = \mp\csc\theta$

$\tan(\theta \pm 90°) = -\cot\theta$

$\cot(\theta \pm 90°) = -\tan\theta$

Complementary Angle Identities

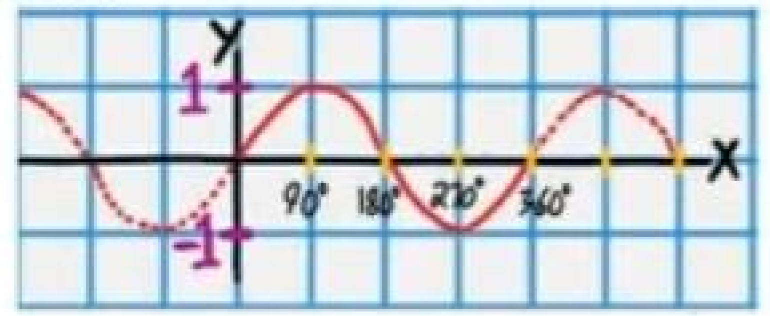
$\sin(90° - \theta) = \cos\theta$ $\csc(90° - \theta) = \sec\theta$

$\cos(90° - \theta) = \sin\theta$ $\sec(90° - \theta) = \csc\theta$

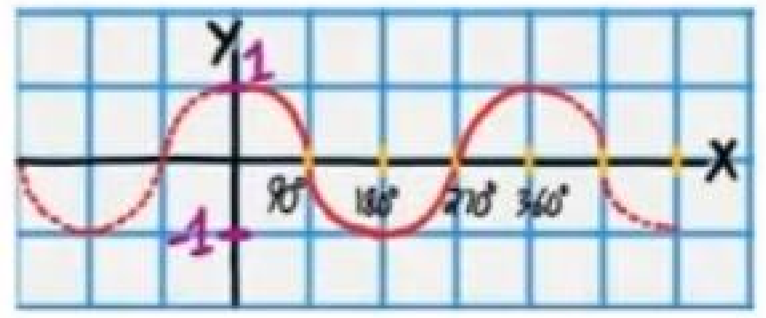
$\tan(90° - \theta) = \cot\theta$

$\cot(90° - \theta) = \tan\theta$

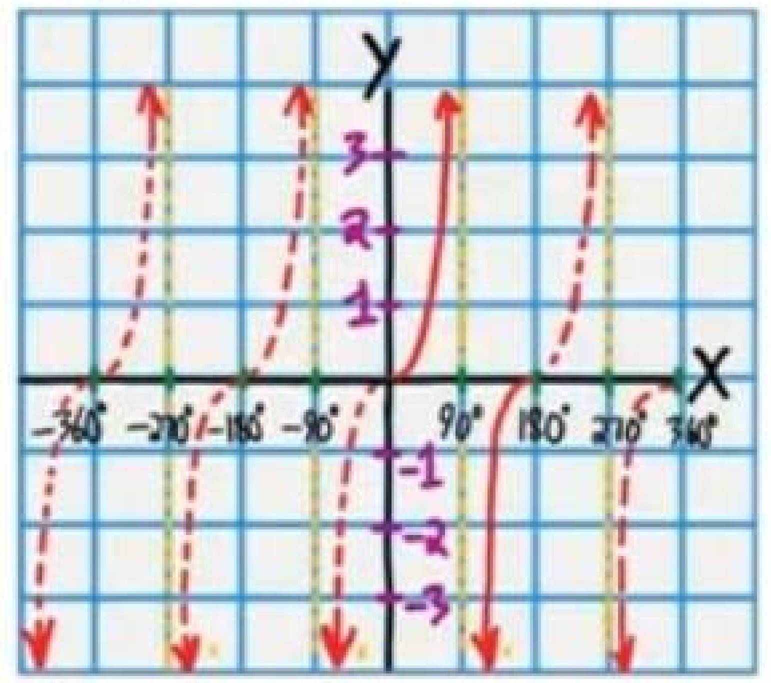
$y = \sin x$



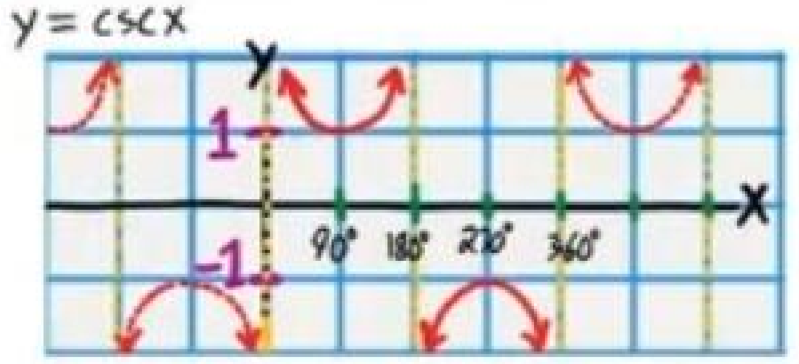
$y = \cos x$



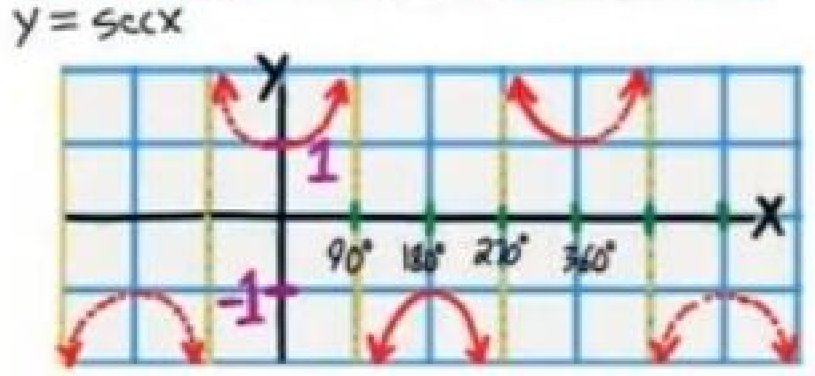
$y = \tan x$



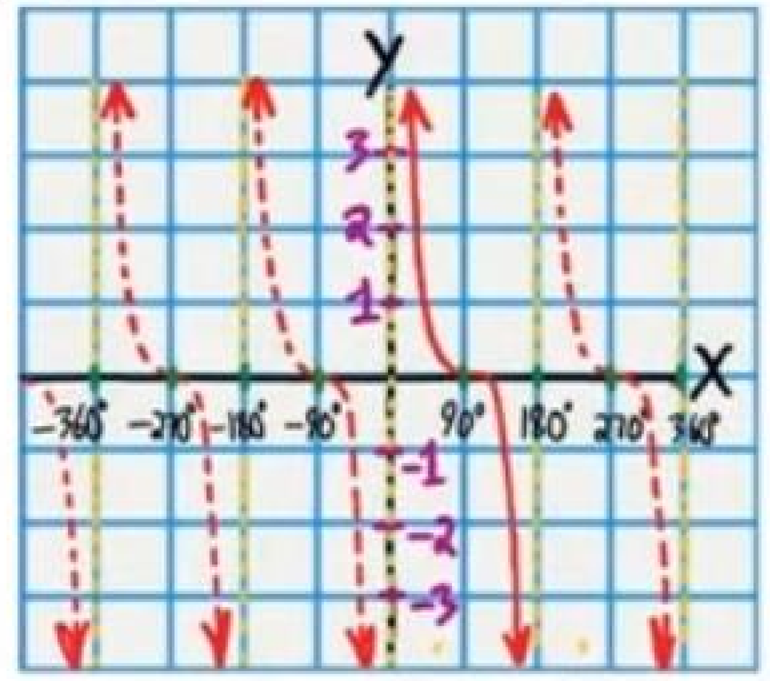
$y = \csc x$



$y = \sec x$



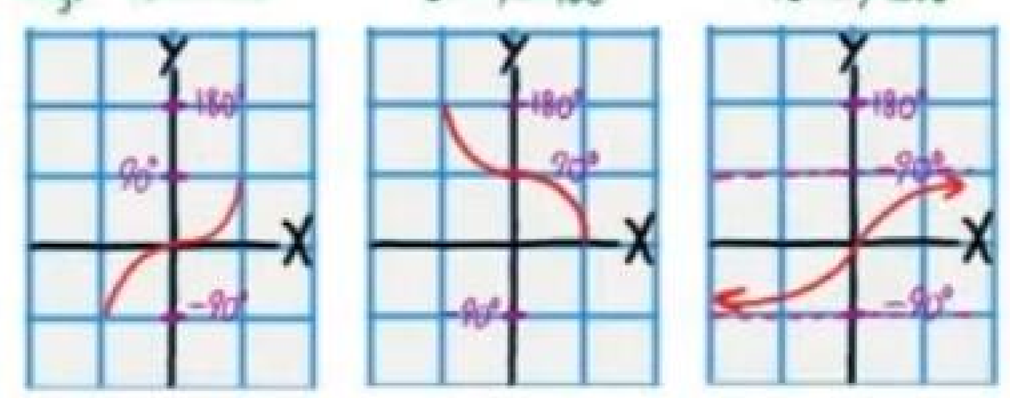
$y = \cot x$



$y = \sin^{-1}x$ Range: $-90° \le y \le 90°$

$y = \cos^{-1}x$ $0° \le y \le 180°$

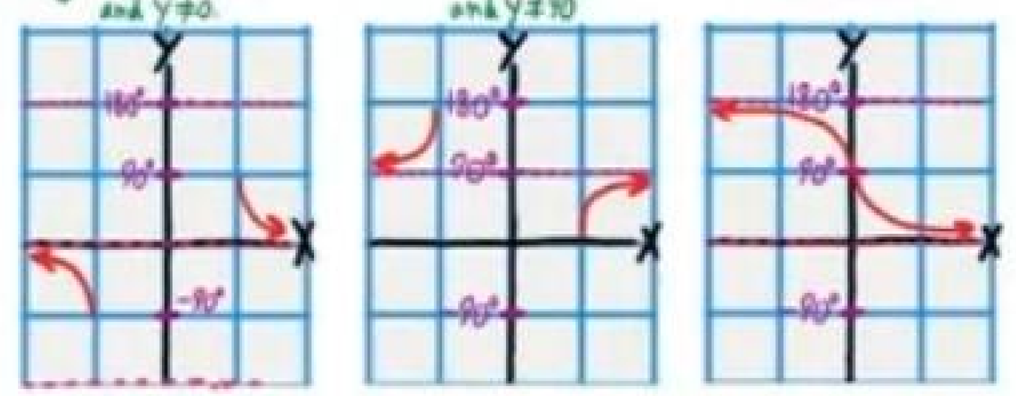
$y = \tan^{-1}x$ $-90° < y < 90°$



$y = \csc^{-1}x$ Range: $-90° \le y \le 90°$ and $y \ne 0$.

$y = \sec^{-1}x$ $0° \le y \le 180°$ and $y \ne 90°$

$y = \cot^{-1}x$ $0° < y < 180°$



Sum to Product Formulas

$\sin A + \sin B = 2\sin\left(\frac{A+B}{2}\right)\cos\left(\frac{A-B}{2}\right)$

$\sin A - \sin B = 2\sin\left(\frac{A-B}{2}\right)\cos\left(\frac{A+B}{2}\right)$

$\cos A + \cos B = 2\cos\left(\frac{A+B}{2}\right)\cos\left(\frac{A-B}{2}\right)$

$\cos A - \cos B = -2\sin\left(\frac{A+B}{2}\right)\sin\left(\frac{A-B}{2}\right)$

Product to Sum Formulas

$\sin A\cos B = \frac{1}{2}[\sin(A+B) + \sin(A-B)]$

$\cos A\sin B = \frac{1}{2}[\sin(A+B) - \sin(A-B)]$

$\cos A\cos B = \frac{1}{2}[\cos(A+B) + \cos(A-B)]$

$\sin A\sin B = \frac{1}{2}[\cos(A-B) - \cos(A+B)]$

Scenarios when Solving Triangles

The Law of Sines

AAS ASA SSA

$\frac{\sin A}{a} = \frac{\sin B}{b} = \frac{\sin C}{c}$

SSA → Ambiguity: i) No triangle ii) One triangle iii) Two triangles

The Law of Cosines

SSS SAS

$c^2 = a^2 + b^2 - 2ab\cos C$

$b^2 = c^2 + a^2 - 2ac\cos B$

$a^2 = b^2 + c^2 - 2bc\cos A$

The Area of an SAS Triangle

The area of a triangle is equal to the length of one side times the length of another side times the sine of the measure of the angle between the two sides divided by two.

$Area = \frac{1}{2}ab\sin C$ $Area = \frac{1}{2}bc\sin A$

$Area = \frac{1}{2}ac\sin B$

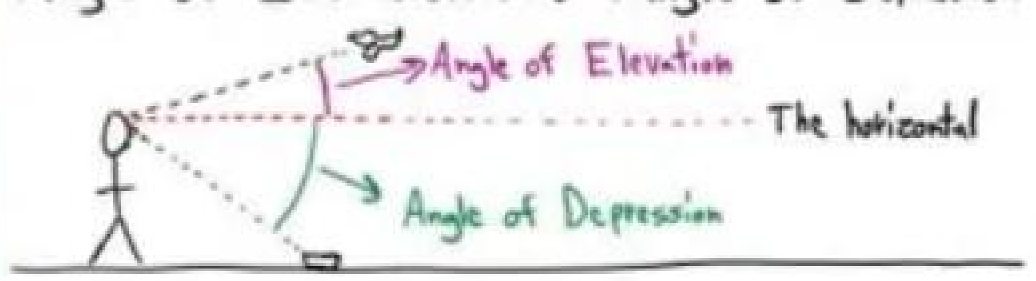
Heron's Formula

If a, b, and c are the side lengths of a triangle, then the area of the triangle is equal to the expression below.

$Area = \sqrt{s(s-a)(s-b)(s-c)}$

where $s = \frac{a+b+c}{2}$

Angle of Elevation and Angle of Depression



Converting from Degrees to Radians

degree measure $\times \frac{\pi \text{ radians}}{180 \text{ degrees}}$ = radian measure

Arc Length

$s = \theta r$ *This formula can be applied with radian units only

The Area of a Sector

$A = \frac{\theta}{2}r^2$ *This formula can be applied with radian units only.

Converting from Radians to Degrees

Radian measure $\times \frac{180 \text{ degrees}}{\pi \text{ radians}}$ = degree measure

If an object moves s units on a circular path in t units of time, then V is defined as the linear velocity according to the equation below.

$V = \frac{s}{t}$

If an object on a circle rotates θ angle units in t units of time, then ω is defined as the angular velocity according to the equation below.

$\omega = \frac{\theta}{t}$

$V = r\omega$ *This formula can be applied with radian units only.

Revolution: A complete rotation.

Revolutions per minute = rpm

1 Revolution = 2π radians = 360°

Name
Date
Course

Practice Final Test (Version 1)

I) This is a cumulative exam.
II) Write your name, the date, and the course title (i.e. Trigonometry).
III) Show all work.
IV) If a problem requires steps to solve, draw a rectangle around your final answer to distinguish it from the other work.
V) You will need a calculator for this test. When pi is required, use the π button on your calculator. Do not use 3.14.
VI) Each problem is worth eight points for a total of 200 points.

Prove that the following equations are true by converting the left side to the right.

① $\dfrac{\tan\theta+\cot\theta}{\tan\theta\cot\theta}=\tan\theta+\cot\theta$

$$\frac{\tan\theta}{\tan\theta\cot\theta}+\frac{\cot\theta}{\tan\theta\cot\theta}=$$

$$\frac{1}{\cot\theta}+\frac{1}{\tan\theta}=$$

$$\tan\theta+\cot\theta=$$

② $\dfrac{\sin^2A+\cos^2A}{1+\tan^2A}=\cos^2A$

$$\frac{1}{\sec^2A}=$$

$$\cos^2A=$$

③ $\sin(W-\alpha)-\sin(W+\alpha)=-2\sin\alpha\cos W$

$$\sin W\cos\alpha-\sin\alpha\cos W-(\sin W\cos\alpha+\sin\alpha\cos W)=$$

$$\cancel{\sin W\cos\alpha}-\sin\alpha\cos W-\cancel{\sin W\cos\alpha}-\sin\alpha\cos W=$$

$$-2\sin\alpha\cos W=$$

④ $\dfrac{\sin(2\beta)}{2\sin\beta}+3\cos\beta=4\cos\beta$

$$\frac{2\sin\beta\cos\beta}{2\sin\beta}+3\cos\beta=$$

$$\cos\beta+3\cos\beta=$$

$$4\cos\beta=$$

⑤ $\csc(-B) + \sec(B-90°) + \cot B = \dfrac{\cos B}{\sin B}$

$-\csc B + \csc B + \dfrac{\cos B}{\sin B} =$

$\dfrac{\cos B}{\sin B} =$ ✓

⑥ Solve if $0 < \beta < 360°$.

$5\sec\beta + 3 = -2\sec\beta + 18$

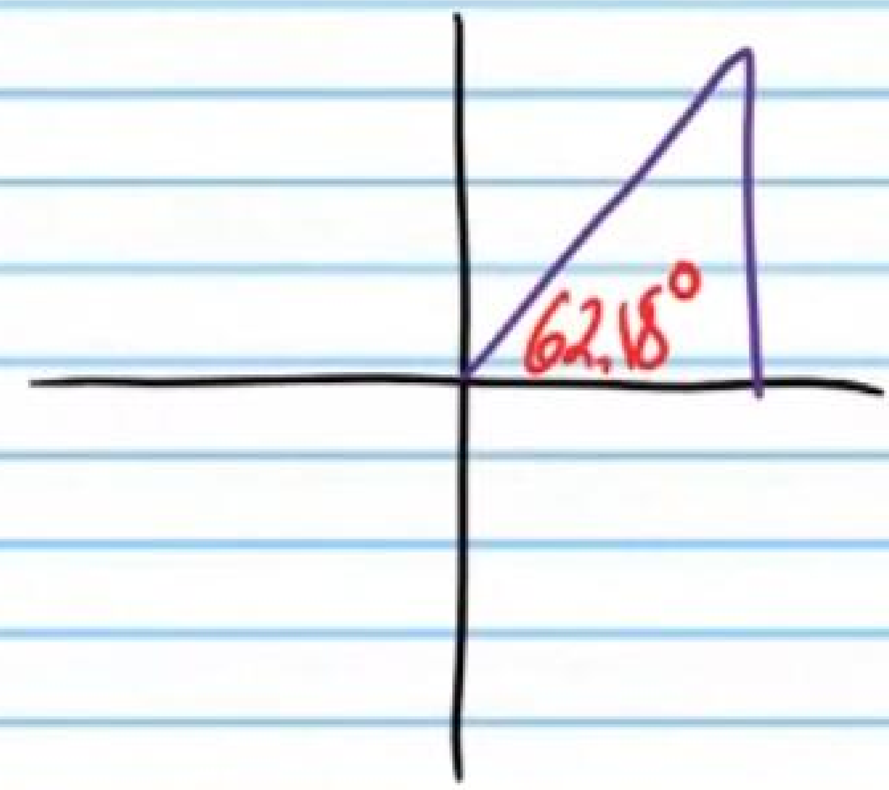
$7\sec\beta = 15$

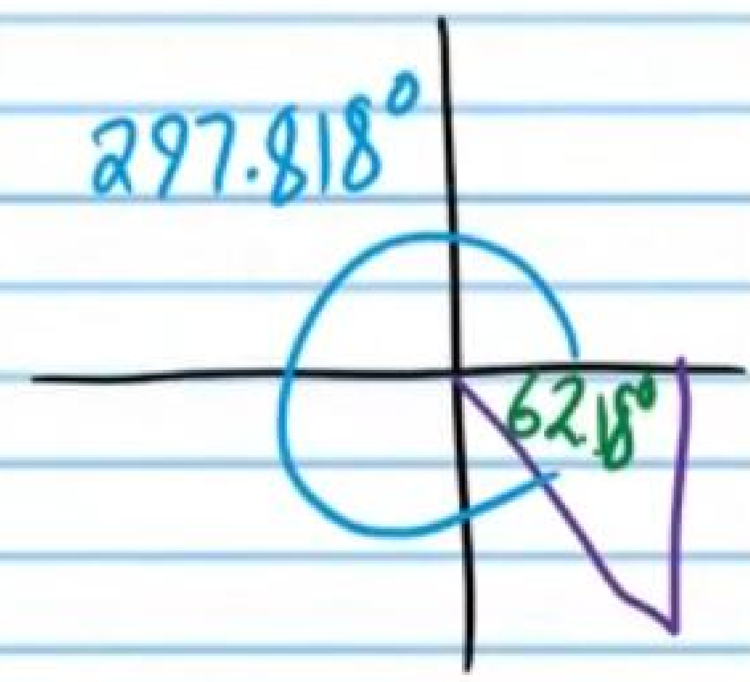
$\sec\beta = \dfrac{15}{7}$

$\dfrac{1}{\cos\beta} = \dfrac{15}{7}$

$\cos\beta = \dfrac{7}{15}$

$\beta = \cos^{-1}\left(\dfrac{7}{15}\right) \approx 62.1818°$

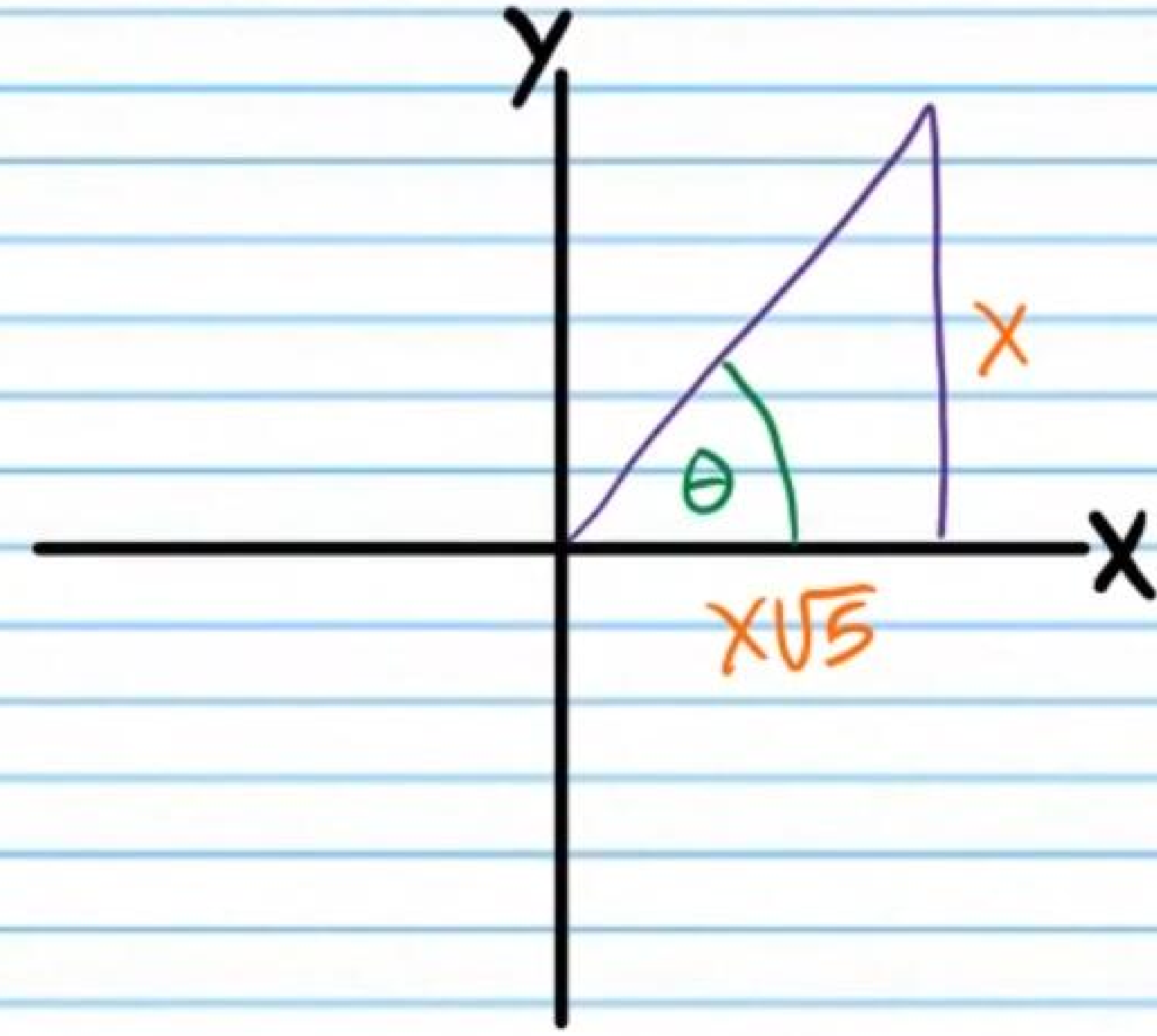


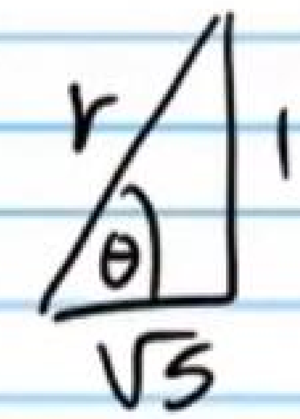


$\boxed{\beta \approx 62.18°,\ 297.818°}$

⑦ If $\tan\theta = \frac{1}{\sqrt{5}}$, and θ terminates in quadrant I, find the values of the five remaining basic trigonometric functions of θ.

Let $x = 1$





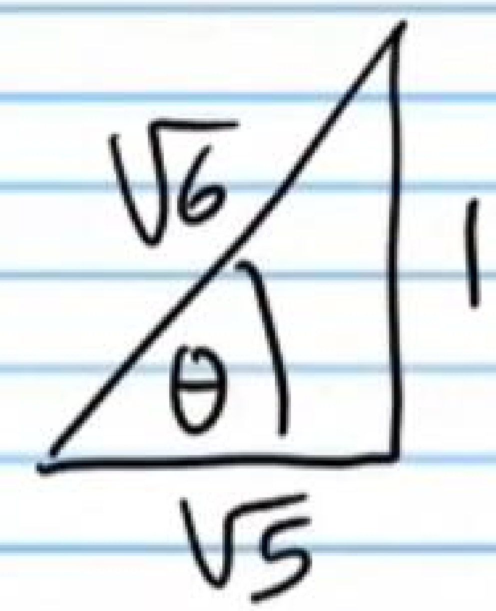
$a^2+b^2=c^2$

$(\sqrt{5})^2+1^2=r^2$

$5+1=r^2$

$r=\pm\sqrt{6}$

$r=\sqrt{6}$



i) $\csc\theta = \dfrac{\sqrt{6}}{1} = \boxed{\sqrt{6}}$

ii) $\sec\theta = \dfrac{\sqrt{6}}{\sqrt{5}} = \boxed{\dfrac{\sqrt{30}}{5}}$

iii) $\cot\theta = \dfrac{\sqrt{5}}{1} = \boxed{\sqrt{5}}$

iv) $\cos\theta = \dfrac{\sqrt{5}}{\sqrt{6}} = \boxed{\dfrac{\sqrt{30}}{6}}$

v) $\tan\theta = \dfrac{1}{\sqrt{5}} = \boxed{\dfrac{\sqrt{5}}{5}}$

⑧ Find the value of the expression below without a calculator.

$\operatorname{arccot}\left(-\frac{3}{\sqrt{3}}\right)$

$\cot^{-1}\left(-\frac{3}{\sqrt{3}}\right)$

$\theta = \cot^{-1}\left(-\frac{3}{\sqrt{3}}\right)$

$\cot\theta = -\frac{3}{\sqrt{3}}$

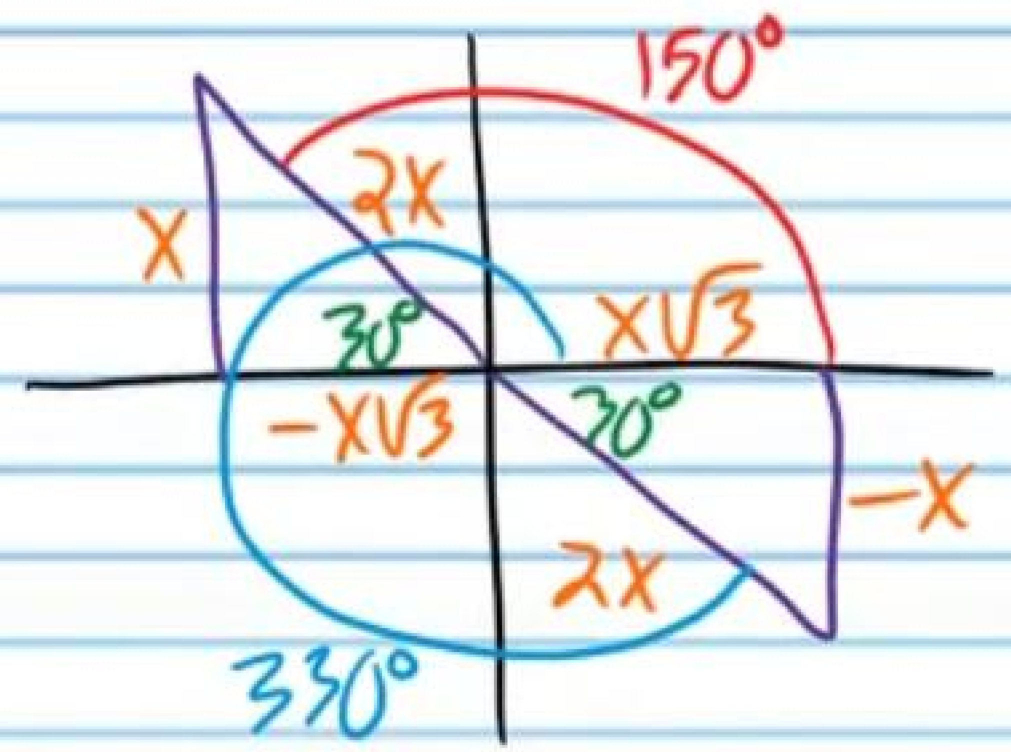
$\cot\theta = -\frac{3}{\sqrt{3}} \cdot \frac{\sqrt{3}}{\sqrt{3}}$

$\cot\theta = -\frac{3\sqrt{3}}{3}$

$\cot\theta = -\sqrt{3}$

$\theta = 150°$

$\operatorname{arccot}\left(-\frac{3}{\sqrt{3}}\right) = \boxed{150°}$



⑨ Solve if $0 \le \theta \le 2\pi$. Write your answer in radian units.

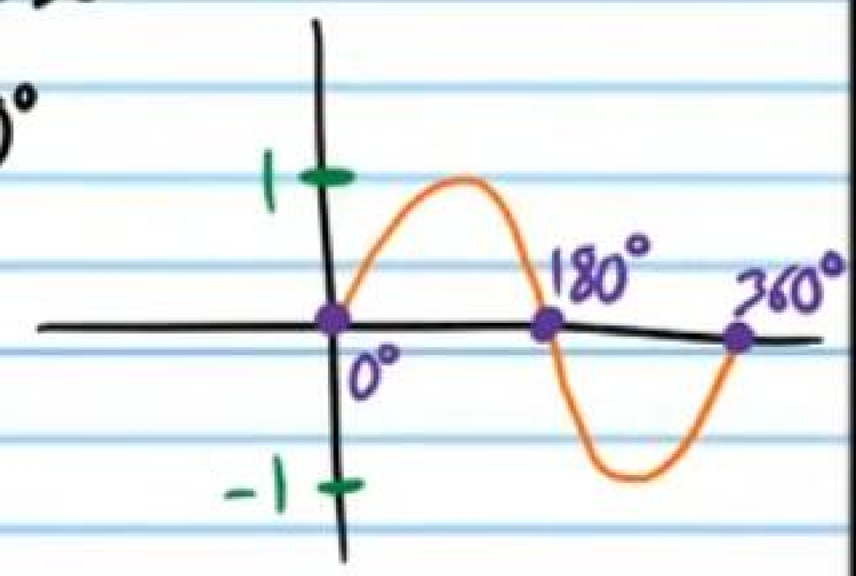
$\sin(3\theta) + 5 = 5$

$\sin(3\theta) = 0$

$3\theta = 0°, 180°, 360°, 540°, 720°, 900°, 1{,}080°$

$\theta = 0°, 60°, 120°, 180°, 240°, 300°, 360°$

$\boxed{\theta = 0, \frac{\pi}{3}, \frac{2\pi}{3}, \pi, \frac{4\pi}{3}, \frac{5\pi}{3}, 2\pi}$



⑩ Find all values of α.

$\cot^2\alpha + 1 = 4 - \csc^2\alpha$

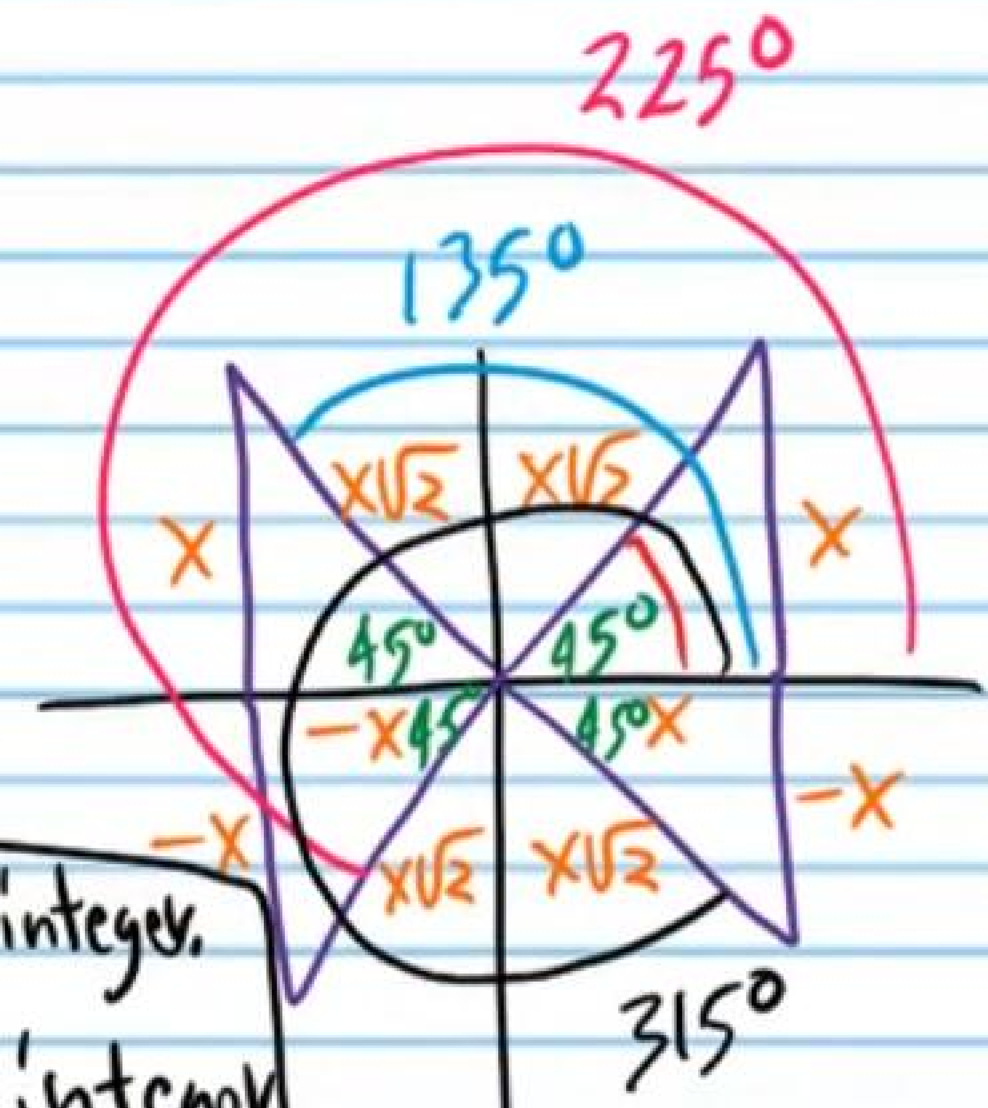
$\csc^2\alpha = 4 - \csc^2\alpha$

$2\csc^2\alpha = 4$

$\csc^2\alpha = 2$

$\csc\alpha = \pm\sqrt{2}$

$\boxed{\begin{array}{l}\alpha = 45° + 180n, \text{ n is any integer.} \\ \alpha = 135° + 180n, \text{ n is any integer}\end{array}}$



(11) Solve if $0 \leq B < 2\pi$. Write your answer in radian units.

$$\cos B - 2 + 2\cos B = \cos(2B) + 6\cos B$$

$$\cos B - 2 + 2\cos B = 2\cos^2 B - 1 + 6\cos B$$

$$3\cos B - 2 = 2\cos^2 B - 1 + 6\cos B$$

$$0 = 2\cos^2 B + 3\cos B + 1$$

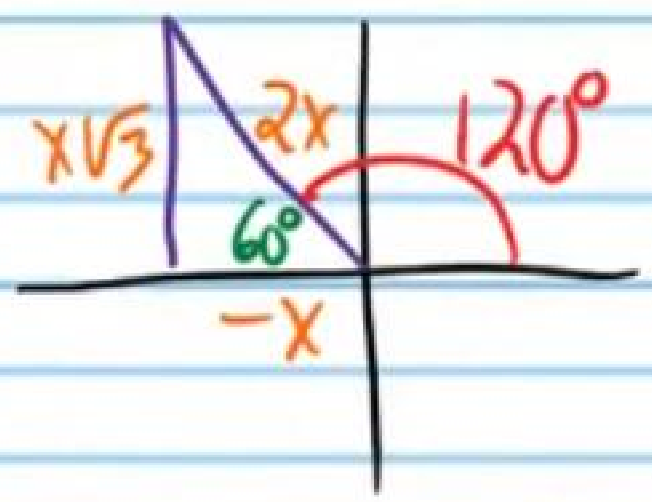
Let $x = \cos B$

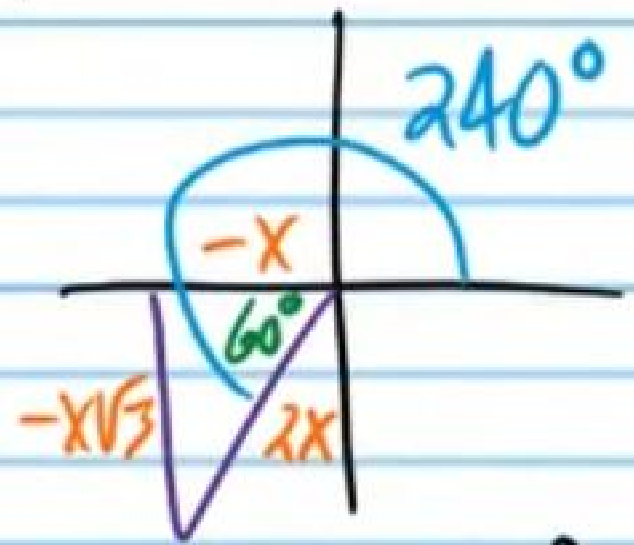
$$0 = 2x^2 + 3x + 1$$

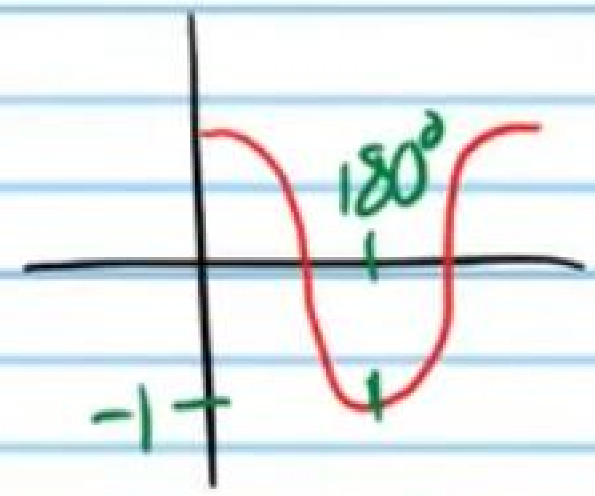
$$0 = (2x+1)(x+1)$$

$$x = -\frac{1}{2} \text{ or } x = -1$$

$$\cos B = -\frac{1}{2} \text{ or } \cos B = -1$$







$B = 120°$ $\quad$ $B = 240°$ $\quad$ $B = 180°$

$$B = 120° \times \frac{\pi}{180°} = \boxed{\frac{2\pi}{3}}, \quad B = 240° \times \frac{\pi}{180°} = \boxed{\frac{4\pi}{3}}, \quad B = \boxed{\pi}$$

Find the values of the following expressions without a calculator.

(12)
$$\sec 345° = \sec(300° + 45°) = \frac{1}{\cos(300° + 45°)}$$

$$= \frac{1}{\cos 300° \cos 45° - \sin 300° \sin 45°}$$

$$= \frac{1}{\left(\frac{1}{2}\right)\left(\frac{1}{\sqrt{2}}\right) - \left(-\frac{\sqrt{3}}{2}\right)\left(\frac{1}{\sqrt{2}}\right)}$$

$$= \frac{1}{\frac{1}{2\sqrt{2}} + \frac{\sqrt{3}}{2\sqrt{2}}}$$

$$= \frac{1}{\frac{\sqrt{2}}{4} + \frac{\sqrt{6}}{4}}$$

$$= \frac{1}{\frac{\sqrt{2} + \sqrt{6}}{4}}$$

$$= \boxed{\frac{4}{\sqrt{2} + \sqrt{6}}}$$

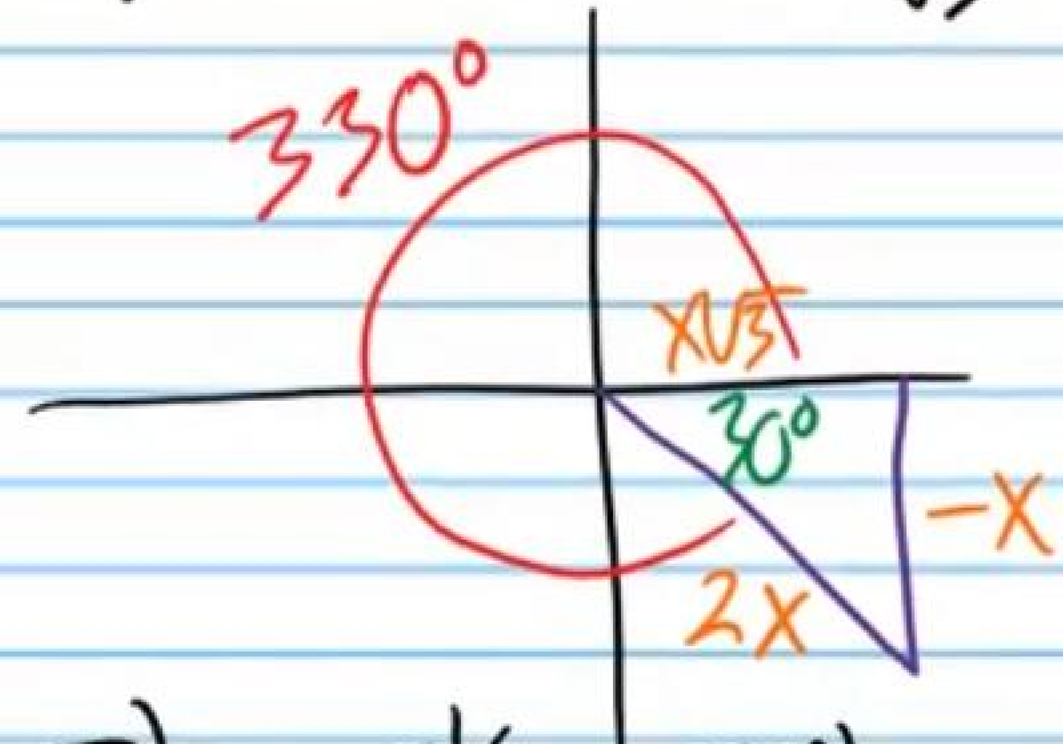
⑬ i) $\sin\left(\sin^{-1}\left(-\frac{1}{\sqrt{2}}\right)\right)$

$-\frac{1}{\sqrt{2}}$

$\boxed{-\frac{\sqrt{2}}{2}}$

ii) $\sec^{-1}(\sec 330^\circ)$

I) $\sec 330^\circ = \frac{2}{\sqrt{3}}$

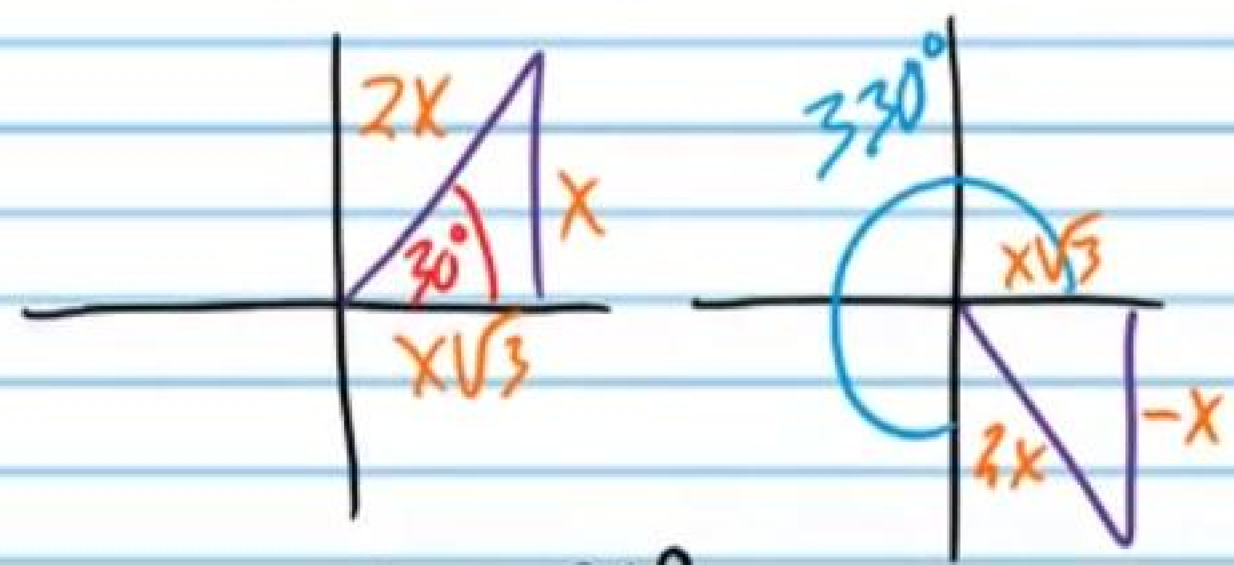


II) $\sec^{-1}(\sec 330^\circ)$

$\sec^{-1}\left(\frac{2}{\sqrt{3}}\right)$

$\theta = \sec^{-1}\left(\frac{2}{\sqrt{3}}\right)$

$\sec\theta = \frac{2}{\sqrt{3}}$



$\theta = 30^\circ$

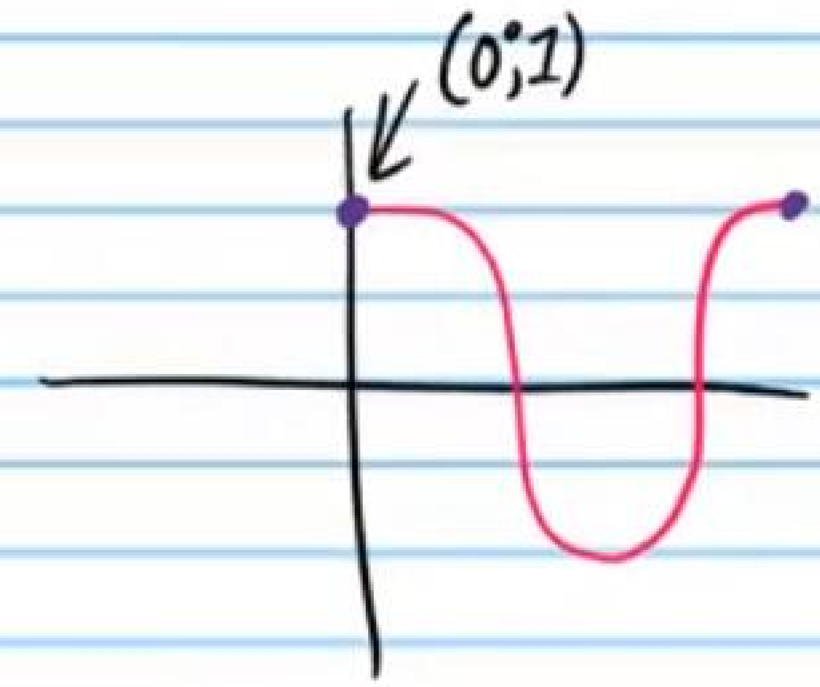
$\sec^{-1}(\sec 330^\circ) = \boxed{30^\circ}$

⑭ $\tan(\cos^{-1} 1)$

I) $\theta = \cos^{-1} 1$

$\cos\theta = 1$

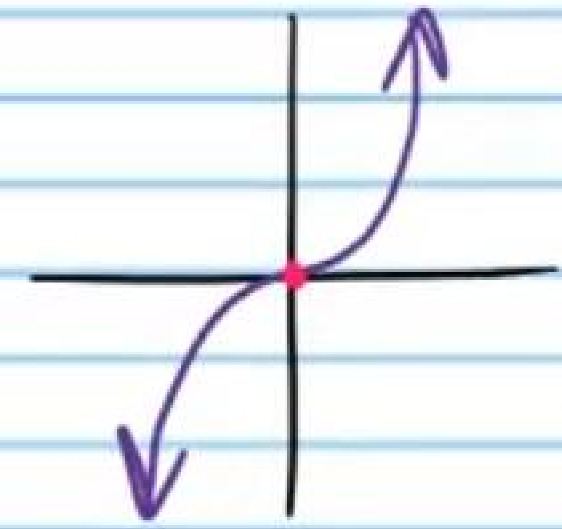
$\theta = 0^\circ$



II) $\tan(\cos^{-1} 1)$

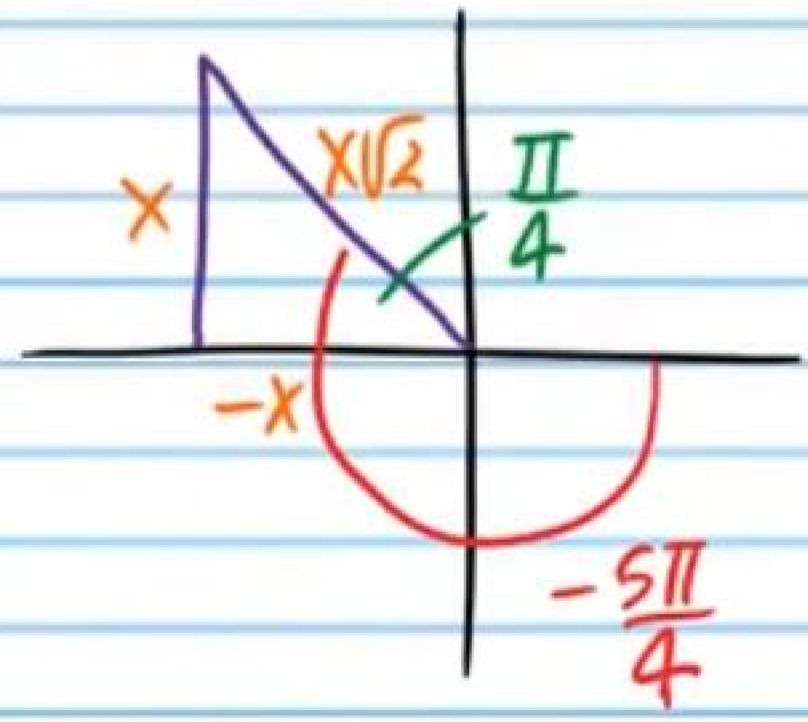
$\tan(0^\circ)$

$\boxed{0}$



(15) $\cot^{-1}\left(\tan\left(-\frac{5\pi}{4}\right)\right)$

I) $\tan\left(\frac{-5\pi}{4}\right) = -1$



II) $\cot^{-1}\left(\tan\left(-\frac{5\pi}{4}\right)\right)$

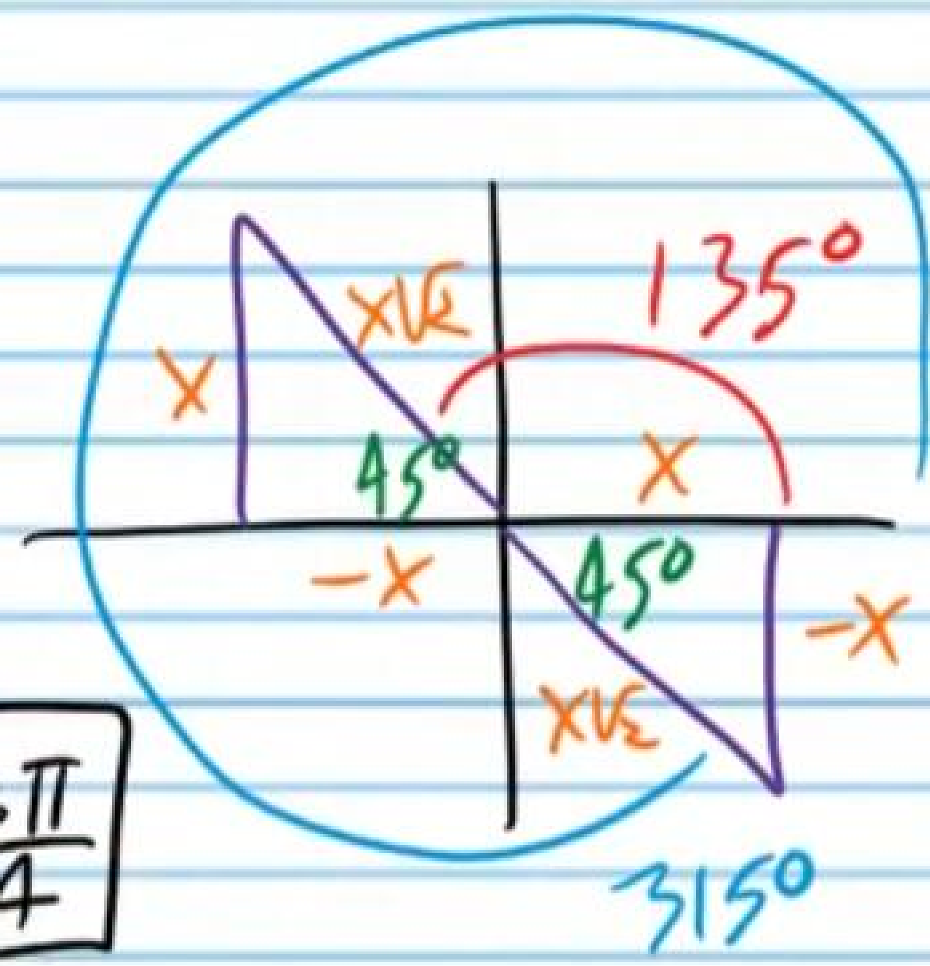
$\cot^{-1}(-1)$

$\theta = \cot^{-1}(-1)$

$\cot\theta = -1$

$\theta = 135°$

$\theta = \frac{3\pi}{4}$



$\cot^{-1}\left(\tan\left(-\frac{5\pi}{4}\right)\right) = \boxed{\frac{3\pi}{4}}$

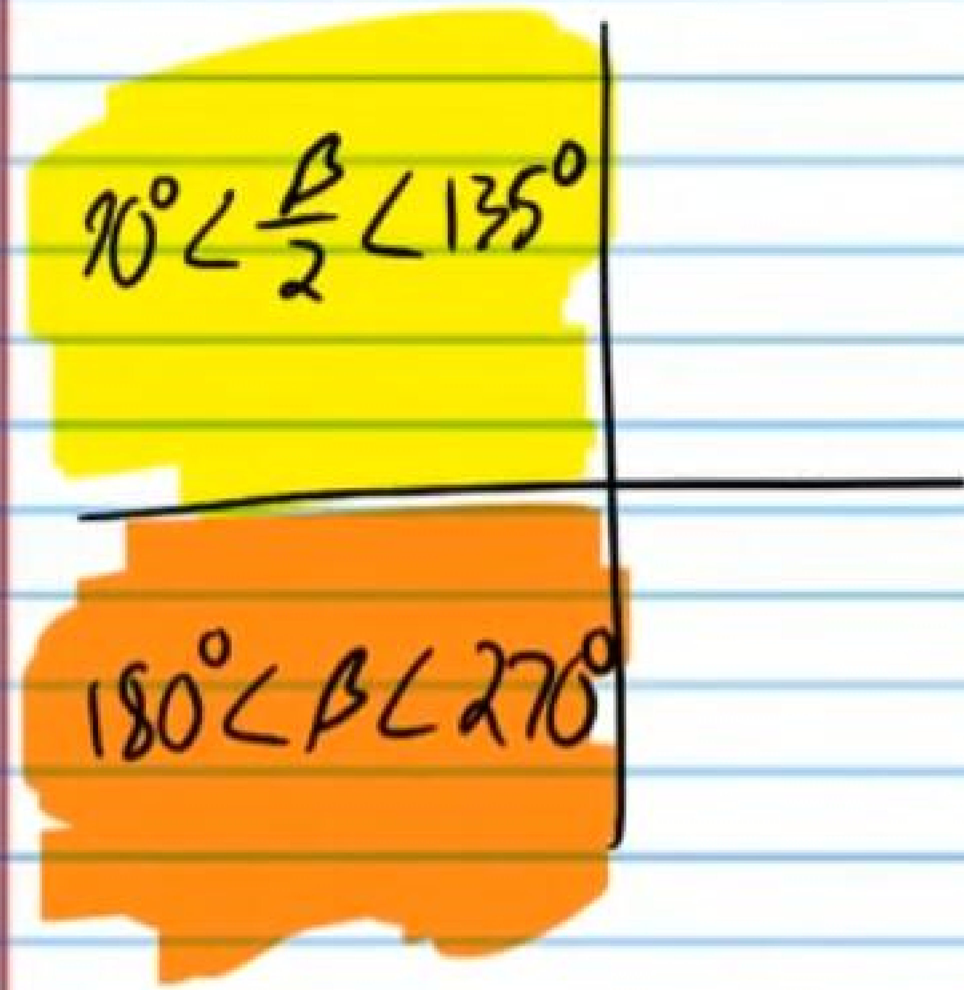
(16) If β is in standard position, and it terminates in QIII ($180° < \beta < 270°$), find the value of each expression below if $\sin\beta = -\frac{\sqrt{11}}{6}$.

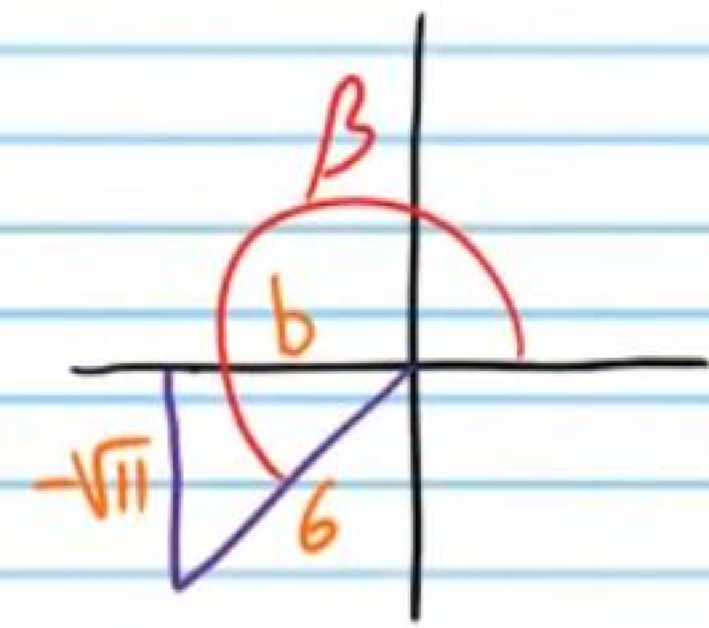
i) $\cos(2\beta) = 1 - 2\sin^2\beta = 1 - 2\left(-\frac{\sqrt{11}}{6}\right)^2 = 1 - 2\left(\frac{11}{36}\right)$

$= 1 - \frac{11}{18}$

$= \boxed{\frac{7}{18}}$

ii) $\cos\left(\frac{\beta}{2}\right) = \pm\sqrt{\frac{1+\cos\beta}{2}} = -\sqrt{\frac{1+\left(-\frac{5}{6}\right)}{2}} = -\sqrt{\frac{\frac{1}{6}}{2}}$





$a^2 + b^2 = c^2$

$(-\sqrt{11})^2 + b^2 = 6^2$

$b = \pm 5$

$b = -5$

$= -\sqrt{\frac{1}{12}}$

$= -\frac{\sqrt{1}}{\sqrt{12}}$

$= -\frac{1}{2\sqrt{3}}$

$= \boxed{-\frac{\sqrt{3}}{6}}$

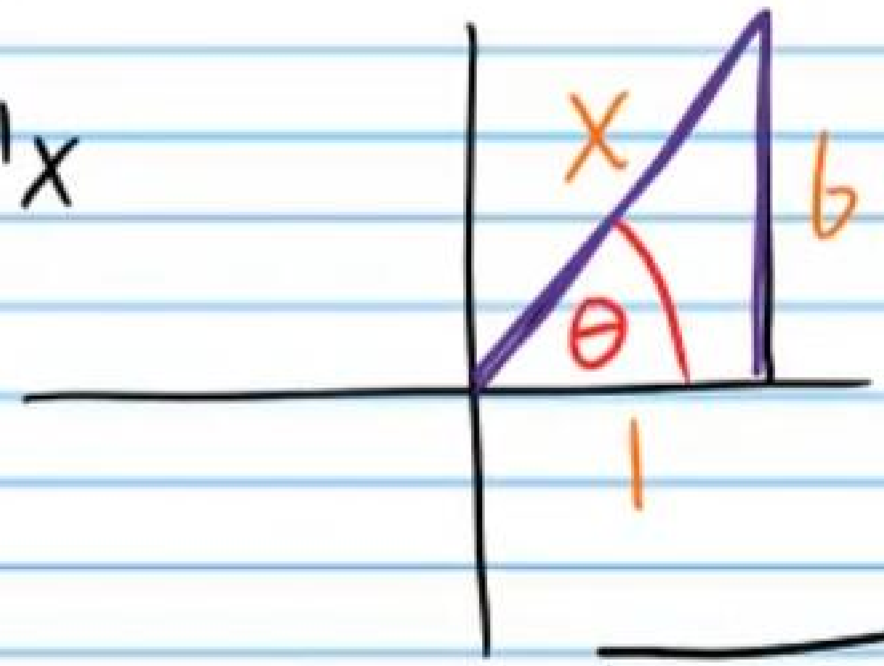
Write an equivalent expression without trigonometric notation. Assume that x is positive.

(17) $\sin(\sec^{-1}x)$

I) $\theta = \sec^{-1}x$

$\sec\theta = x$

$\sec\theta = \frac{x}{1}$



$a^2+b^2=c^2$

$1^2+b^2=x^2$

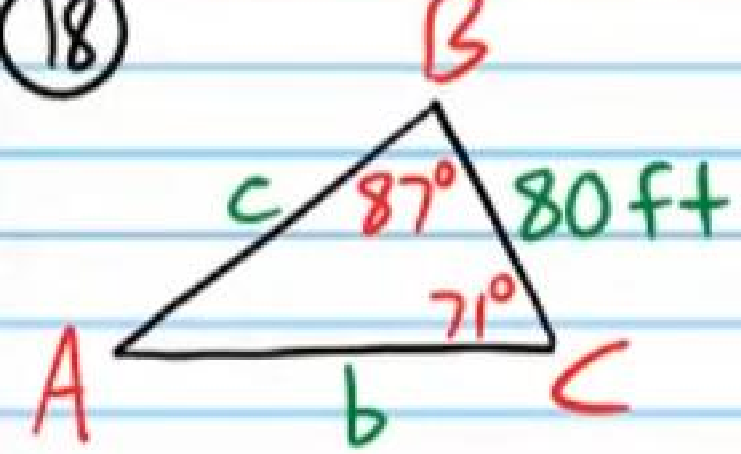
$b=\pm\sqrt{x^2-1}$

$b=\sqrt{x^2-1}$

II) $\sin(\sec^{-1}x) = \sin\theta = \boxed{\frac{\sqrt{x^2-1}}{x}}$

Find the missing side lengths and angle measures of each triangle. In other words, solve each triangle. A triangle may not be possible for some problems.

(18)



I) Find $m\angle A$.

$A+B+C=180°$

$A+87°+71°=180°$

$\boxed{A=22°}$

II) Find c.

$\frac{\sin C}{c} = \frac{\sin A}{a}$

$\frac{\sin 71°}{c} = \frac{\sin 22°}{80}$

$c = \frac{80\sin 71°}{\sin 22°}$

$\boxed{c \approx 201.92 \text{ ft}}$

III) Find b.

$\frac{\sin A}{a} = \frac{\sin B}{b}$

$\frac{\sin 22°}{80} = \frac{\sin 87°}{b}$

$b = \frac{80\sin 87°}{\sin 22°}$

$\boxed{b \approx 213.26 \text{ ft}}$

(19) $a = 20\text{mm},\ b = 17\text{mm},\ B = 46^\circ$

I) Find $m\angle A$.

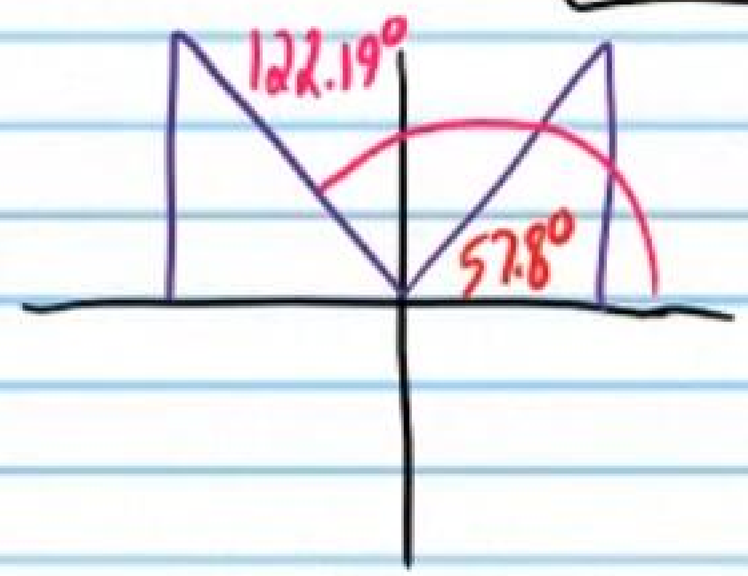
$$\frac{\sin A}{a} = \frac{\sin B}{b}$$

$$\frac{\sin A}{20} = \frac{\sin 46^\circ}{17}$$

$$\sin A = \frac{20\sin 46^\circ}{17}$$

$$\sin A \approx 0.84628\ldots$$

$$A_1 \approx \boxed{57.8095^\circ\ldots}$$



$$A_1 + B < 180^\circ$$

$$57.8^\circ + 46^\circ < 180^\circ$$

$$A_2 + B < 180^\circ$$

$$122.19^\circ + 46 < 180^\circ$$

Two Δ's Possible

II) Find $m\angle C_1$.

$$A_1 + B + C_1 = 180^\circ$$

$$57.8 + 46^\circ + C_1 \approx 180^\circ$$

$$\boxed{C_1 \approx 76.19^\circ}$$

III) Find c_1

$$\frac{\sin C_1}{c_1} = \frac{\sin B}{b}$$

$$\frac{\sin 76.1904^\circ}{c_1} \approx \frac{\sin 46^\circ}{17}$$

$$c_1 \approx \frac{17\sin 76.1904^\circ}{\sin 46^\circ}$$

$$\boxed{c_1 \approx 22.95\text{mm}}$$

IV) Find $m\angle A_2$.

$$\boxed{A_2 \approx 122.19^\circ}$$

V) Find $m\angle C_2$.

$$A_2 + B + C_2 = 180^\circ$$

$$122.19^\circ + 46^\circ + C_2 \approx 180^\circ$$

$$\boxed{C_2 \approx 11.8^\circ\ldots}$$

VI) Find c_2.

$$\frac{\sin C_2}{c_2} = \frac{\sin B}{b}$$

$$\frac{\sin 11.8^\circ\ldots}{c_2} \approx \frac{\sin 46^\circ}{17}$$

$$c_2 \approx \frac{17\sin 11.8^\circ}{\sin 46^\circ}$$

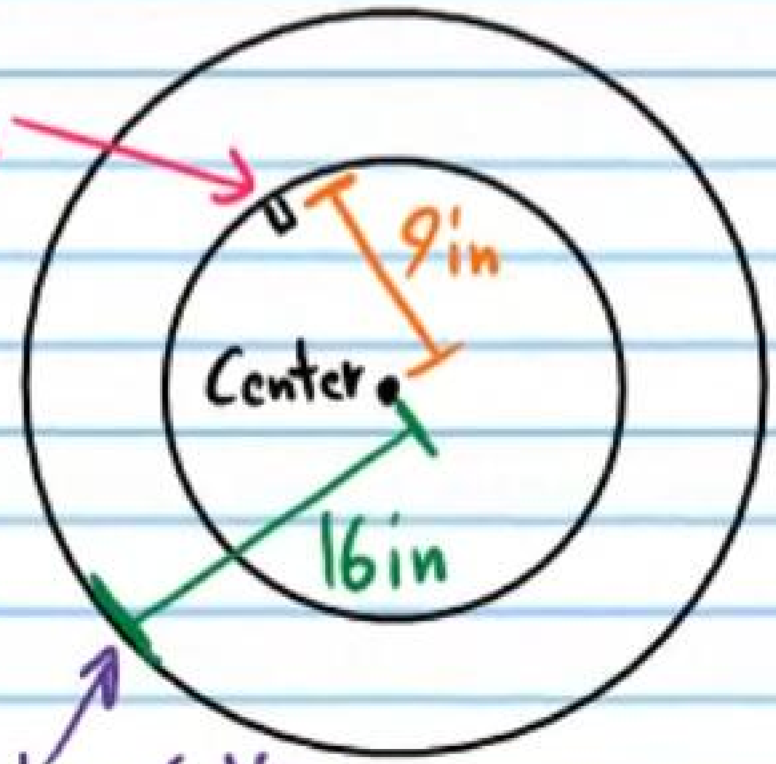
$$\boxed{c_2 \approx 4.837\text{mm}}$$

(20) A valve stem on a rotating wheel is 9 inches from the center of the wheel and moves at 35 inches per second. What is the angular velocity of the wheel? How fast does the far edge of the tire move in inches per second if the radius of the wheel is 16 inches?

I) $V_1 = r_1 w$

$35 = 9w$

$\boxed{w = \frac{35}{9}\ \text{rad/s}}$



II) $V_2 = r_2 w = 16\left(\frac{35}{9}\right)$

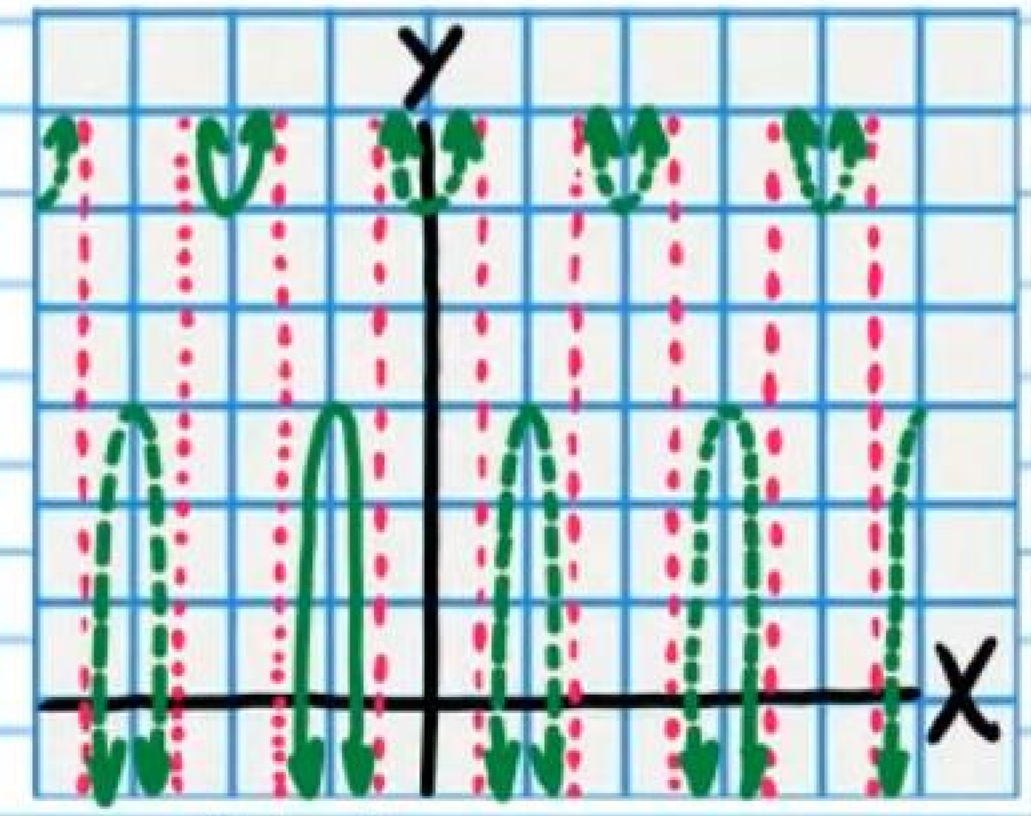
$= \boxed{62.\overline{2}\ \text{in/s}}$

(21) Write an equation corresponding to the graph below. Use the cosecant function rather than the secant function. Each horizontal unit on the graph is 90°. Each vertical unit is one.

Midline Equation: $y = 4$

$B = 4$

A: $A = 1$



Omega: $P = \frac{360^\circ}{w}$

$180^\circ = \frac{360^\circ}{w}$

$w = 2$

Phase Shift:

$PS = \frac{\Phi}{w}$

$-225^\circ = \frac{\Phi}{2}$

$-450^\circ = \Phi$

$y = A\csc(wx - \Phi) + B$

$y = (1)\csc(2x - (-450^\circ)) + 4$

$\boxed{y = \csc(2x + 450^\circ) + 4}$

(22) Find the period, amplitude, phase shift, and the equation of the midline of the graph of the equation below. Then graph the equation.

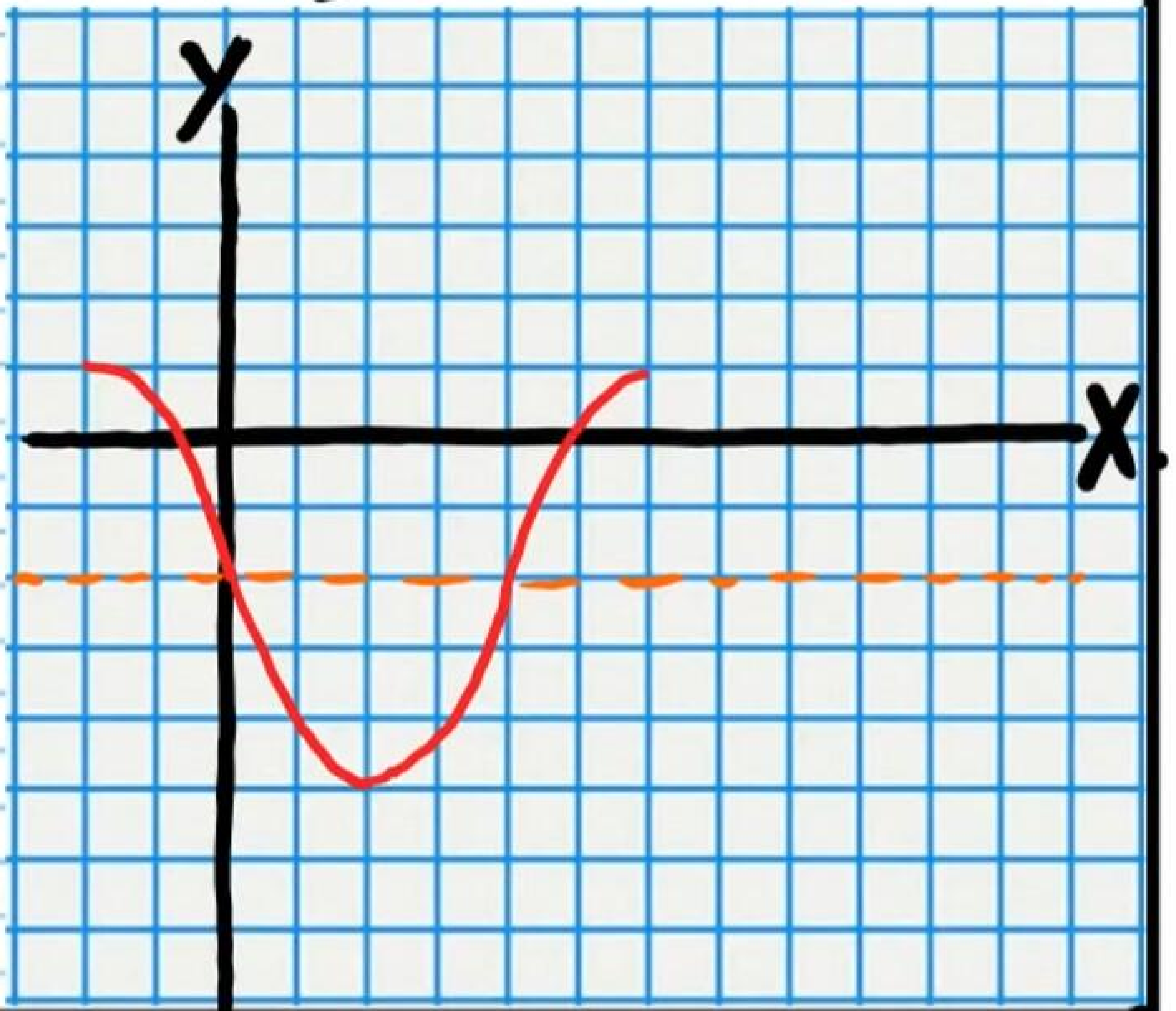
$$y = 3\cos\left(\frac{x}{2} + 90^\circ\right) - 2$$

Midline Equation: $\boxed{y = -2}$

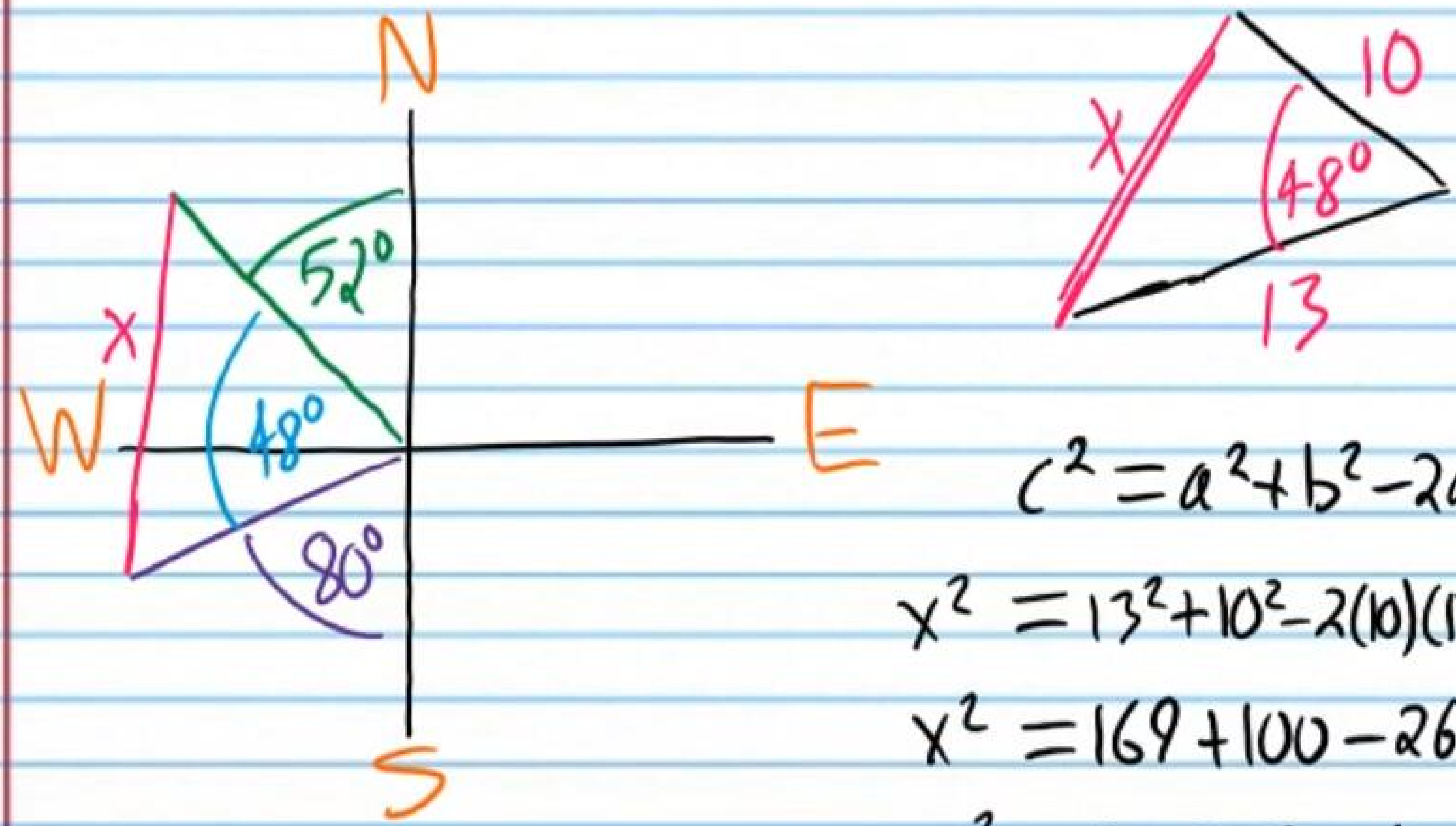
Period: $P = \frac{360^\circ}{\omega} = \frac{360^\circ}{\frac{1}{2}} = \boxed{720^\circ}$

Phase Shift: $PS = \frac{\Phi}{\omega} = \frac{-90^\circ}{\frac{1}{2}} = \boxed{-180^\circ}$

Amplitude: $A = |3| = \boxed{3}$



(23) A map shows that a person in the town of Belfast must hike 10 kilometers on a course 52 degrees west of north (N52°W) to arrive at Huntington Beach. If a person wants to hike from Belfast to the town of Ashland, he must walk 13 kilometers on a course 80 degrees west of south (S80°W). How far apart are Huntington Beach and the town of Ashland?



$$c^2 = a^2 + b^2 - 2ab\cos C$$

$$x^2 = 13^2 + 10^2 - 2(10)(13)\cos 48^\circ$$

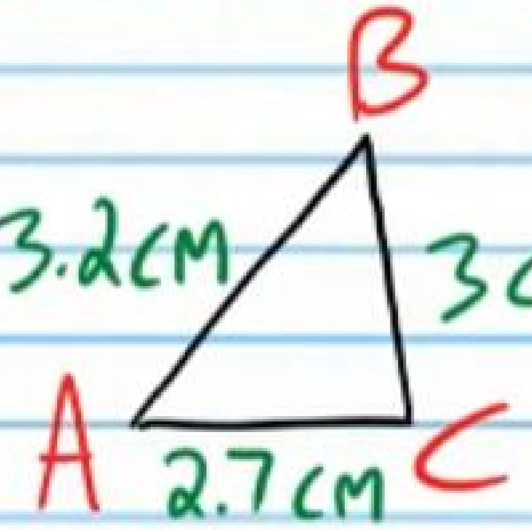
$$x^2 = 169 + 100 - 260\cos 48^\circ$$

$$x^2 \approx 269 - 173.9739\ldots$$

$$x^2 \approx 95.026\ldots$$

$$\boxed{x \approx 9.748 \text{ KM}}$$

(24) Find the area of the triangle below. Round to the nearest thousandth.



I) $s = \frac{a+b+c}{2} = \frac{3.2+3+2.7}{2}$

$= 4.45$

II) $\text{Area} = \sqrt{s(s-a)(s-b)(s-c)}$

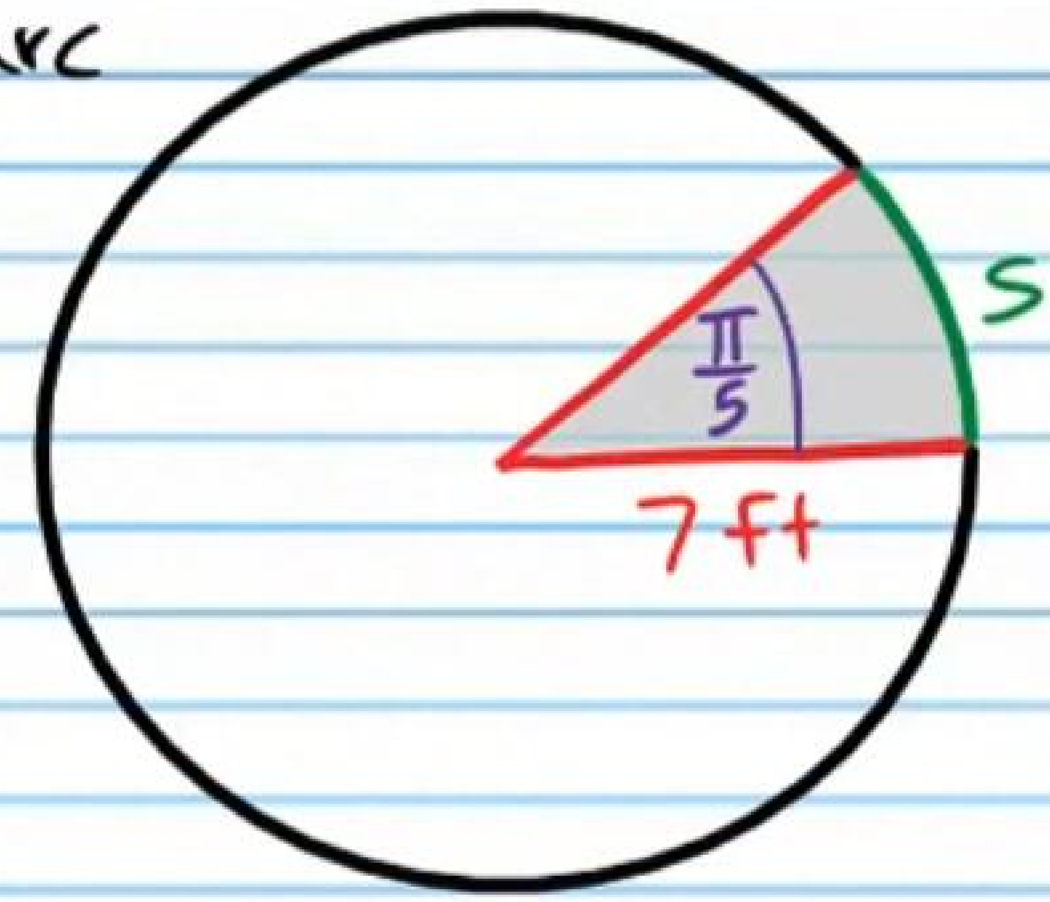
$= \sqrt{4.45(4.45-3.2)(4.45-3)(4.45-2.7)}$

$= \sqrt{4.45(1.25)(1.45)(1.75)}$

$\approx \sqrt{14.1148}$

$\approx \boxed{3.757\ cm^2}$

(25) i) Find the length of arc "s" on the circle below.



$s = r\theta = 7\left(\frac{\pi}{5}\right) = \boxed{\frac{7\pi}{5}\ ft}$

ii) Find the area of the shaded region in the circle.

$\text{Area} = \frac{\theta}{2}r^2 = \frac{\frac{\pi}{5}}{2}7^2 = \frac{\pi}{10}49 = \boxed{\frac{49\pi}{10}\ ft^2}$